STATISTICS MASTER'S COMPREHENSIVE EXAMINATION

23 March 2024

Instructions: Attempt any five out of the following six questions. Specify which five you would like to have graded. Students are allowed calculators, laptops, books and notes. Internet use and cell phones are prohibited. Students are allowed to use laptops for viewing electronic copies of textbooks, but for nothing else.

Problem 1. Scientists at Rutgers have discovered a new one-dimensional world where density of an observation X is centered at a point a and declines symmetrically as the distance from a increases in either direction according to this

$$f(X) = (\lambda/2)e^{-\lambda|X-a|}$$

where $\lambda > 0$ and $|X - a|$ is the absolute value of the difference between X and a.

The following independent observations from this distribution have been observed:

-7 -5 -3 1 2 6 10

(1) FIRST ANSWER THIS: For a given fixed value of λ what is the MLE for a?

Answer:

$L(\lambda, a) = \prod_{i=1}^{N} (\lambda/2)e^{-\lambda|X_i-a|}$ *so* $\ell(\lambda, a) = \sum_{i=1}^{N} \log(\lambda) - \log(2) - \lambda/2|X_i - a|$.

Taking the derivative with respect to a:

$$\frac{\partial \ell}{\partial a} = -\lambda \sum_{i=1}^{N} [I(a \geq x_i) - I(a < x_i)].$$

Setting the derivative to zero:

$$\sum_{i=1}^{N} I(a < x_i) = \sum_{i=1}^{N} I(a > x_i).$$

The a that satisfies the above equation is the sample median. So the MLE of a is the 1.

(2) SECOND ANSWER THIS Find the MLE for λ at the value of a you derived above. If you are not able to derive a value for a then let $a = 0$ to answer this question.

Answer:

Take the derivative of the log-likelihood with respect to λ:

$$\frac{\partial \ell(\lambda, 1)}{\partial \lambda} = \frac{n}{\lambda} - \sum_{i=1}^{n} |x_i - 1|.$$

Setting to zero and solving for λ:

$$\widehat{\lambda} = \frac{n}{\sum_{i=1}^{n} |x_i - 1|}.$$

We have:

$$\widehat{\lambda} = \frac{7}{|-7-1| + |-5-1| + |-3-1| + |1-1| + |2-1| + |6-1| + |10-1|} = \frac{7}{33}$$

(3) A new one-dimensional world has been discovered where the density of X again is $f(X) = (\lambda/2)e^{-\lambda|X-a|}$ but the values of λ and a are different than for the previous problem. Furthermore, this time it is known that $a = 2$. Scientists were able to measure 20 independent observations, but this time they could only determine if X was ≤ 15; 5 of the 20 observations were > 15 and 15 were ≤ 15. What is the MLE for λ?

Answer:

We have

$$\begin{aligned} P(X > 15) &= \int_{15}^{\infty} \frac{\lambda}{2} e^{-\lambda|x-2|}\, dx \\ &= \int_{13}^{\infty} \frac{\lambda}{2} e^{-\lambda z}\, dz \end{aligned}$$

where $z = x - 2$ (this is always greater than zero so no absolute value sign is needed).

$$\begin{aligned} P(X > 15) &= \left(-\frac{1}{2}e^{-\lambda z}\right)\Big|_{13}^{\infty} \\ &= \frac{1}{2}e^{-13\lambda} \end{aligned}$$

Then, the probability of 5 of 20 observations being greater than 15, and 15 observations being less than 15, is:

$$L(\lambda) = \binom{20}{5}\left[\frac{1}{2}e^{-13\lambda}\right]^5 \left[1 - \frac{1}{2}e^{-13\lambda}\right]^{15}$$

The log-likelihood is then:

$$\ell(\lambda) = \log\binom{20}{5} + 5\left[\log(0.5) - 13\lambda\right] + 15\log\left(1 - \frac{1}{2}e^{-13\lambda}\right).$$

Taking the derivative:

$$\frac{\partial \ell}{\partial \lambda} = -5 \times 13 + \frac{(15 \times 13 \times 0.5)e^{-13\lambda}}{1 - 1/2e^{-13\lambda}}.$$

Setting the derivative to zero:

$$(13 \times 5)\left[-1 + \frac{(3 \times 0.5)e^{-13\lambda}}{1 - 1/2e^{-13\lambda}}\right] = 0$$

$$\frac{3}{2}e^{-13\lambda} = 1 - \frac{1}{2}e^{-13\lambda}$$
$$2e^{-13\lambda} = 1$$
$$\lambda = -\frac{1}{13}\log(0.5).$$

So $\lambda \approx 0.0533$.

Problem 2. An experiment was conducted to determine the effectiveness of a new drug in lowering cholesterol. Ten patients were included in the study. Prior to treatment, cholesterol was measured for each patient. The patient was administered the new drug for a six-month period. Cholesterol on each patient was measured again. The data from this study are given below:

Patient	Baseline Cholesterol	Final Cholesterol
1	210	190
2	230	210
3	200	210
4	200	180
5	220	220
6	230	200
7	250	230
8	190	210
9	240	200
10	230	210

(1) Determine if the drug has resulted in a reduction from baseline in cholesterol. In selecting a statistical method you may assume the data are normally distributed. Be sure to show the null and alternative hypotheses.
Answer: H_0*: Mean reduction in cholesterol is 0.*
H_a*: One-sided Per the question. Mean reduction is* > 0 *which means Mean Final -Baseline will be* < 0.
Paired t-test

Patient	*Baseline*	*Final*	*Change*
Patient	*Cholesterol*	*Cholesterol*	
1	*210*	*190*	*-20*
2	*230*	*210*	*-20*
3	*200*	*210*	*+10*
4	*200*	*180*	*-20*
5	*220*	*220*	*0*
6	*230*	*200*	*-30*
7	*250*	*230*	*-20*
8	*190*	*210*	*+20*
9	*240*	*200*	*-40*
10	*230*	*210*	*-20*

The mean change is

$$\frac{-20-20+10-20+0-30-20+20-40-20}{10}=-14.$$

The sample variance is:

$$\begin{aligned}&\left[(-20--14)^2+(-20--14)^2+(+10--14)^2+(-20--14)^2+(+0--14)^2\right.\\&\quad\left.+(-30--14)^2+(-20--14)^2+(+20--14)^2+(-40--14)^2+(-20--14)^2)\right]/9\\&=\frac{36+36+576+36+196+256+36+1156+676+36}{9}\\&=337.7778.\end{aligned}$$

The standard deviation is $\sqrt{337.7778}=18.37873$.

The standard deviation of mean is is $18.37873/\sqrt{10}=5.811865$, *and the test statistic is* $T=-14/5.811865=-2.4<1.833$ *which is the fifth percentile of t distribution with 9 df.*

Therefore we reject H_0.

(2) It is now discovered that an 11th patient randomly recruited from the same population under the same selection criteria as the other 10 patients and was given the same drug for 6 months. Give a 95% confidence interval for that patient's final cholesterol.
Answer: The mean is the sum of the column marked "Final Cholesterol" above, divided by 10, or 206.

The variance is the sum of the final values minus 206, squared, and divided by $10-1$, *or 204.4444.*

The standard deviation is $\sqrt{204.4444}=14.29841$. *The 95% CI is*

$$\bar{x}\pm t_{0.975,9df}S\sqrt{1+1/n}$$

or $206\pm 2.262\times 14.29841\sqrt{1+1/10}$, *which is* $(172.0784, 239.9216)$

(3) This study was designed to determine whether the drug causes a decrease in choesterol. When designing this study the experimenter made one important mistake. Indicate what change you feel is necessary to make this study acceptable to careful review.
Answer: Needed an untreated control group for comparison to separate specific effects due to drug from other effects due to time.

Problem 3. We want to generate the single continuous random variable X by using either density function $f(x)=2x$, $0<x<1$ or the density function $g(x)=3x^2$, $0<x<1$. To decide which density to use we generate a Bernoulli random variable Y such that

$$P(Y=1)=\frac{1}{2}=1-P(Y=0).$$

If $Y=1$ we use density $f(x)$. If $Y=0$ we use density $g(x)$.

(1) Find the cumulative distribution function for X.

Answer: The cdf for $f(x)$ *is* $F(x) = x^2$ *for* $0 < x < 1$*. The cdf for* $g(x)$ *is* $G(x) = x^3$ *for* $0 < x < 1$*. Thus the overall cdf for* X *is* $(x^2 + x^3)/2$ *for* $0 < x < 1$*.*

(2) Find $P(Y = 1|X \leq 1/2)$.
Answer: $P(Y = 1|X \leq 1/2) = P(X \leq 1/2|Y = 1)P(Y = 1)/P(X \leq 1/2)$*. From (a) this equals* $(1/2)^2(1/2)/([(1/2)^2+(1/2)^3]/2) = (1/4)(1/2)(16/3) = 2/3$*.*

(3) Find $Cov(X, Y)$.
Answer:

$$E[X|Y = 1] = \int_0^1 xf(x)\ dx = 2/3$$

$$E[X|Y = 0] = \int_0^1 xg(x)\ dx = 3/4$$

Thus $E[X] = (1/2)(2/3) + (1/2)(3/4) = 17/24$ *and* $E[Y] = 1/2$*. Also*

$$\begin{aligned} E[XY] &= E[E[XY|Y]] \\ &= E[YE[X|Y]] \\ &= P(Y = 1)E[X|Y = 1] \\ &= 1/3 \end{aligned}$$

Finally,

$$\begin{aligned} Cov(X, Y) &= E[XY] - E[X]E[Y] \\ &= 1/3 - 17/24 \times 1/2 \\ &= 1/48. \end{aligned}$$

Problem 4. We have <u>**ONE**</u> continuous random variable X with density function $f(x|\theta) = \begin{cases} 2x/\theta^2 & \text{if } 0 < x < \theta \\ 0 & \text{otherwise.} \end{cases}$.

(1) Find $E_\theta X$ and $Var_\theta X$.
Answer: $E_\theta X = 2\theta/3$*,* $E_\theta X^2 = \theta^2/2$*. Thus* $Var_\theta(X) = \theta^2/18$*.*

(2) Based on X we wish to estimate θ when the loss function is $L(\theta, a) = (a - \theta)^2$. Consider estimators of the form $\delta_c(x) = cx$. Find the best (smallest Mean Squared Error) estimator of this form. That is, find the optimal choice for c.
Answer: MSE= $E_\theta(cX–\theta)^2 = c^2E_\theta X^2–2^c\theta E_\theta X + \theta^2$ *. Taking the derivative wrt c and setting = 0 gives* $c = \theta E_\theta X/E_\theta X^2 = 4/3$*. Thus the best estimator of this type is* $4X/3$*.*

(3) Is the estimator you found in part (a) an unbiased estimator of θ ?
Answer: $E_\theta[4X/3] = 4(2\theta/3)/3 = 8\theta/9$*. Not unbiased.*

(4) Find an unbiased estimator of $\theta(1–\theta)$.
Answer: From (a): $E_\theta[(3/2)X–2X^2] = \theta(1–\theta)$*.*

Problem 5. In an experiment testing the effect of a toxic substance, 1500 experimental insects were divided at random into six groups of 250 each. The insects in each group were exposed to a fixed dose of the toxic substance. A day later, each insect was observed. Death from exposure was recorded 1, and survival was recorded 0. The results are shown below where X_j denotes the dose level (on a logarithmic scale) administered to the insects in group j and Y_j denotes the number of insects that died out of the 250 in the group (n_j=250).

j	1	2	3	4	5	6
X_j	1	2	3	4	5	6
n_j	250	250	250	250	250	250
Y_j	28	53	93	126	172	194

The logistic regression model is assumed to be appropriate, i.e., the estimated proportions $p_j = \frac{Y_j}{n_j}$ against X_j.

(1) Assume that the maximum likelihood estimates of β_0 and β_1, the intercept and slope, in the logistic regression model is $\hat{\beta} = -2.64$, $\hat{\beta}_1 = 0.67$. State the fitted response function.
Answer: $\pi = P(Y_i = 1) = \frac{\exp(-2.64+0.67x_i)}{1+\exp(-2.64+0.67x_i)} = \frac{1}{1+\exp(2.64-0.67x_i)}$.

(2) What is the estimated probability that an insect dies when the dose level is at $X = 3.5$?
Answer: $\pi = \frac{1}{1+\exp(2.64-0.67\times3.5)} = 0.4268$.

(3) What is the estimated median lethal dose – that is, the dose for which 50 percent of the experimental insects are expected to die?
Answer: $\pi = \frac{1}{1+\exp(2.64-0.67\times x_1)} = 0.5$, *so* $x_i = 3.922$.

Problem 6. The `kid_iq` data has been analyzed, where the kid's IQ test score `kid_score` is treated as the output (Y), and the mom's IQ test score `mom_iq` (X_1) and the mom's age `mom_age` (X_2) at the birth of the child are predictors. Answer the questions based on the `R` output given below.

```
> m1=lm(kid_score~mom_iq+mom_age,data=kidiq)
> summary(m1)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 17.59625    9.08397   1.937   0.0534 .
mom_iq       0.60357    0.05874     ???      ___
mom_age      0.38813    0.32620     ???      ___
---

Residual standard error: 18.26 on 431 degrees of freedom
Multiple R-squared:  0.2036,    Adjusted R-squared:  ___
F-statistic:  ??? on 2 and 431 DF,  p-value: < 2.2e-16
```

```
> m2=lm(kid_score~mom_iq,data=kidiq)
> anova(m2,m1)
Analysis of Variance Table

Model 1: kid_score ~ mom_iq
Model 2: kid_score ~ mom_iq + mom_age
  Res.Df    RSS Df Sum of Sq      F Pr(>F)
1    432 144137
2    431 143665  A      B    ??? 0.2348
```

(1) What is the sample size of this data?
Answer: 432= n- number of parameters in model 1. So $432 = n - 2$ *and* $n = 434$.

(2) Report the fitted model `m1`.
Answer: kidscore=17.596+0.604momiq+0.388momage

(3) For model `m1`, what is the least square estimate of β_1 (the coefficient of X_1)? Give an interpretation of this coefficient.
Answer: $\hat{\beta}_1 = 0.6036$.

(4) For model `m1`, test whether the coefficient of X_1 is significantly different from zero, using level 5%.
Answer: $H_0 : \hat{\beta}_1 = 0$ *v.s.* $H_1 : \hat{\beta}_1 \neq 0$.
$T = |\frac{0.6036-0}{0.0587}| = 10.28 > 1.965$ *which is* t_{431} *with* 5% *level. Thus, we reject* H_0.

(5) Calculate the F statistic for testing the whole regression model, for `m1`.
Answer: $F = \frac{(TSS-RSS)/p}{RSS/n-p-1}$. *Now* $RSS = 143665$, $p = 2$, $n - p - 1 = 431$, *and* $R^2 = 1 - \frac{RSS}{TSS} = 0.2036$, *we have* $TSS = 180393$. $F = 55.092$.

(6) What is the t statistic for testing whether β_2 is zero, under `m1`?
Answer: $H_0 : \hat{\beta}_2 = 0$ $H_1 : \hat{\beta}_2 \neq 0$.
t-statistic: $0.38813/0.32620 = 1.189853$.

(7) State H_0 and H_1 of the ANOVA test, comparing `m1` and `m2`.
Answer: $H_0 : \hat{\beta}_2 = 0$ *v.s.* $H_1 : \hat{\beta}_2 \neq 0$.

(8) In the ANOVA table, what are the values for "A" and "B"? What is the F-value of the ANOVA test? What are the degrees of freedom of the corresponding F distribution? What is the p-value? What is your conclusion, with level 5%?
Answer: $A = 432 - 431 = 1$ *and* $B = 144137 - 143665$.
$F = \frac{B/A}{143665/431} \sim F_{1,431}$. *The p value is* 0.2348, *so we do not reject* H_0.